BreakfastWise

Market Requirements Document (MRD)

Breakfast-guest forecasting for small and medium-sized hotels

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Status: Working draft for team and mentor review | **Version:** 2.0 | **Date:** 24 October 2026 | **Important:** market assumptions and success metrics are not verified results; they are marked ASSUMPTION or proposed target.

1. Executive Summary

BreakfastWise is a proposed dashboard that forecasts how many guests will eat breakfast, using expected occupancy and past breakfast attendance. The first version solves the **headcount problem only**; converting guests into item quantities is a later step.

Core need: kitchen teams at small and medium-sized hotels need a practical, written way to estimate breakfast demand so they can reduce avoidable overproduction and shortages without relying only on one person's memory. **Proposed MVP:** record expected occupancy and actual breakfast guests; validate and store them; forecast guests with the calculation shown; warn when history is too short; compare forecast with actual.

Pitch: For head chefs of small hotels, who plan breakfast from memory and guesswork, BreakfastWise forecasts the number of breakfast guests from expected occupancy and past records, so the kitchen starts from a consistent, checkable number. Unlike memory alone, it shows its calculation, so the chef can verify every forecast by hand.

MRD status: target segment, persona priority and performance targets are proposed assumptions, not validated market findings or achieved performance.

2. Target Segment and Users

Initial segment — ASSUMPTION pending validation: independent small and medium-sized hotels serving about 50–200 breakfast guests per day. Not yet supported by interviews or market data.

PERSONA	GOALS	PAIN POINTS TO VALIDATE	CURRENT WORKAROUND
Primary: Head chef / kitchen manager	Prepare suitable quantities; avoid shortages and excess	Estimate lives in one person's head; unreliable when numbers change, the hotel is busy or the chef is away	Memory and experience; previous-day counts; paper notes
Secondary: Hotel manager	See how well forecasts match actual attendance	No single view of occupancy and breakfast attendance	PMS reports, spreadsheets, manual summaries
Supporting: Kitchen staff	Record actual breakfast guests after service	Manual recording may be inconsistent	Paper registers, verbal updates

3. Problem, Need and Rationale

Problem: breakfast is prepared before the final guest count is known. Underestimating causes shortages; overestimating leaves food that is discarded. **Typical morning (ASSUMPTION):** the head chef decides quantities around 5:00 AM from expected occupancy, yesterday's experience and judgement; service starts around 7:00 AM. Some days fewer guests arrive and cooked food goes unused; on others, popular items (idli, dosa, pongal, poori) run out.

Two problems, one solved first. *Wrong headcount* (guests differ from expectation) is solved first. *Wrong dish mix* (items run out although the total is right) needs dish-level history and is a later step. Headcount comes first because a forecast closer to actual guests than the chef's unaided estimate gives a better starting point for planning.

Need (ASSUMPTION): avoid money wasted on excess food; under-preparation also hurts guest satisfaction. The size of the cost is unknown and will be measured in a pilot by recording discarded food (kg) and its preparation cost. **Evidence status:** no hotel interviews, waste baseline or real dataset exist yet. The MVP does not claim to reduce waste or save money.

4. Existing Alternatives and Opportunity Gap

ALTERNATIVE	HOW IT MAY BE USED	POTENTIAL GAP TO VALIDATE
Chef's memory and experience (main)	Estimate from yesterday's count, occupancy and judgement	Not written down; fails when numbers change suddenly, the hotel is busy or the chef is away; no record of past error
Previous-day / week counts; Excel; paper logs	Quick reference or manual tracking of attendance	Does not account for changing occupancy; analysis and comparison are manual
Hotel PMS; inventory / food-cost tools	Reservations, occupancy, stock and recipe cost	Breakfast attendance forecasting may be missing; verify each hotel's setup

Differentiator: a written guest forecast whose calculation is shown on screen, so a chef can check it by hand. Competitor capabilities have not been researched in depth.

5. Product Scope and Key Capabilities

CAPABILITY	DESCRIPTION	STATUS
Data entry and records	Expected occupancy by date; actual breakfast guests after service; validate dates and non-negative values	MVP core
Validation and storage	Reject missing, negative, invalid and duplicate records; keep valid records after restart	MVP core
Demand forecasting	Predicted guests = expected occupancy × historical consumption rate; calculation shown	MVP core
Data-sufficiency warning	Under 7 valid records: insufficient; 7–29: low confidence; 30+: reliable (proposed thresholds)	MVP if time allows
Forecast vs. actual, history, dashboard	Compare forecast with actual guests; show history and trends by date	MVP if time allows
Item quantities; wastage tracking; ML models	Guests to idli/dosa/pongal/poori quantities; discarded-food tracking; Random Forest / Gradient Boosting	Later steps; not built

Demo flow: (1) select date → (2) enter expected occupancy → (3) review forecast and its calculation, or an insufficient-data notice → (4) change occupancy and watch the forecast recalculate → (5) record actual guests → (6) compare forecast vs. actual and history, with synthetic data labelled.

6. Data Plan

No hotel partner or real records exist, so the demo uses **60 days of synthetic history** (about 50–200 guests/day, weekday/weekend variation, a few unusually busy or quiet days). Every dashboard view and forecast carries this label:

> **SYNTHETIC DEMO DATA — NOT REAL HOTEL DATA. Forecasts use simulated records. Accuracy has not been validated using real hotel data.**

Honest limits: the data is generated from an assumed rate, so the demo proves the workflow runs, not real-world accuracy. The script, seed and rate are documented in the README. The 7 and 30 record thresholds are system rules, not proof of accuracy. The forecast is a transparent formula, not a trained model; comparing it with Random Forest and Gradient Boosting is a future step once enough real data exists.

7. Proposed Success Metrics (Targets, Not Results)

METRIC	PROPOSED TARGET	HOW TO EVALUATE
Guest forecasting error	MAE ≤ 10 guests/day	Predictions vs. actual on later held-out dates; on synthetic data this shows the workflow, not real accuracy
Data validation	≥ 95% of invalid test records flagged	Missing, negative, invalid-date and invalid-number records
Response time; workflow tests	Under 3 s locally; 100% of test cases pass	Repeated local runs; the agreed test suite

Deferred to a pilot: forecast vs. the chef's unaided estimate, item portions within ±15%, and wastage reduction of 15% or more; all need real data and a controlled baseline.

8. Assumptions, Unknowns and Risks

ITEM	ASSUMPTION / RISK	VALIDATION OR MITIGATION
A1 Segment fit	Hotels with 50–200 breakfast guests/day have a real planning problem and would use a lightweight tool	Interview chefs and managers; observe workflows
A2 Data access	Hotels can provide consistent occupancy and attendance history	Ask pilot hotels what they record
A3 Current practice	Chefs estimate mainly from memory and previous counts	Confirm with one real chef, manager or mess supervisor
Risk 1	Chef does not trust the forecast	Screen shows its working so the chef can check it by hand; test in a pilot
Risk 2	Staff do not record actual guests, so nothing can be compared	Keep entry quick; measure completion in a pilot
Risk 3	Expected occupancy is unreliable	Staff update occupancy before breakfast and the forecast recalculates

9. Out of Scope, Open Decisions and Next Steps

Out of scope: item-level quantities and wastage tracking; ML models; integration with every PMS; autonomous purchasing; guaranteed savings; automatic retraining; event-data feeds; multi-hotel benchmarking.

OPEN DECISION / NEXT STEP	PROPOSED OWNER	OUTPUT NEEDED
Talk to one real chef, hotel manager or mess supervisor	Business Analyst	One real quote on how breakfast quantities are decided
Trim PRD to three Must-Haves; write epics and stories	Business Analyst + QA	PRD, backlog with acceptance criteria
Document synthetic data generation and test protocol	Developer + QA	README, time-based evaluation plan, test cases

Review / approval: Business Analyst: Pending | Project Mentor: Pending | Team review: Pending Date: _____